Quantum Field Theory
Part I: Motivation and Foundation

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Abstract

This document contains derivations of equations from [1, Part I: Motivation and Foundation].

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1.1 Path Integral Formulation

Theorem 1 (Some useful integrals)**.**

$$\int_{-\infty}^{\infty} dx e^{-\frac{1}{2}x^2} = \sqrt{2\pi} \tag{1}$$

$$\int_{-\infty}^{\infty} dx e^{-\frac{1}{2}ax^2} = \left(\frac{2\pi}{a}\right)^{\frac{1}{2}} \tag{2}$$

$$\int_{-\infty}^{\infty} dx\, e^{-\frac{a}{2}x^2+Jx} = \left(\frac{2\pi}{a}\right)^{\frac{1}{2}} e^{\frac{J^2}{2a}} \tag{3}$$

$$\int_{-\infty}^{\infty} dx\, e^{-\frac{a}{2}x^2+iJx} = \left(\frac{2\pi}{a}\right)^{\frac{1}{2}} e^{-\frac{J^2}{2a}} \tag{4}$$

Proof.

$$\begin{aligned}
\int_{-\infty}^{\infty} dx e^{-\frac{1}{2}x^2}\big)\big(\int_{-\infty}^{\infty} dy e^{-\frac{1}{2}y^2}\big) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} dx dy e^{-\frac{1}{2}(x^2+y^2)} \\
&= \int_0^{2\pi} d\theta \int_0^{\infty} dr\ r\ e^{\frac{-r^2}{2}}, \text{ substituting } x = r\cos\theta, y = r\sin\theta \\
&= -2\pi \int_0^{\infty} du e^{-u}\ , \text{ substituting } u = \frac{-r^2}{2} \\
&= -2\pi e^{-u}\Big|_0^{\infty} \\
&= 2\pi
\end{aligned}$$

Taking square roots of both side, we get (1).

$$\begin{aligned}
\int_{-\infty}^{\infty} dx e^{-\frac{1}{2}ax^2} &= \frac{1}{\sqrt{a}}\int_{-\infty}^{\infty} dx' e^{-\frac{1}{2}x'^2}, \text{ substituting } x' = \sqrt{a}x \\
&= \left(\frac{2\pi}{a}\right)^{\frac{1}{2}}, \text{ from (1), which gives (2)}
\end{aligned}$$

$$\begin{aligned}
\int_{-\infty}^{\infty} dx\, e^{-\frac{a}{2}x^2+Jx} &= \int_{-\infty}^{\infty} dx\, e^{-\frac{a}{2}\left[x^2-2\left(\frac{J}{a}\right)x+\left(\frac{J}{a}\right)^2\right]} e^{\frac{a}{2}\left(\frac{J}{a}\right)^2}, \text{ completing the square} \\
&= e^{\frac{J^2}{2a}}\int_{-\infty}^{\infty} dx e^{-\frac{a}{2}\left[x-\frac{J}{a}\right]^2}, \text{ now substitute } x' = x - \frac{J}{a} \\
&= e^{\frac{J^2}{2a}}\int_{-\infty}^{\infty} dx' e^{-\frac{a}{2}x'^2}
\end{aligned}$$

Substituting (2) gives (3). Replacing J by iJ gives (4). □

Theorem 2 (Feynman Path Integral–Exercise I.2.1 - eq(5))**.** *If the Hamiltonian is given by*

$$H = \frac{\hat{p}^2}{2m} + V(\hat{q}), \textit{ and the corresponding Lagrangian is} \tag{5}$$

$$L = \frac{\hat{p}^2}{2m} - V(\hat{q}), \textit{ then} \tag{6}$$

$$\langle q_F|H|q_I\rangle = \int Dq(t) e^{i\int_0^T dt L}, \textit{ where} \tag{7}$$

$$\int Dq(t) \triangleq \lim_{N\to\infty} \left(\frac{-im}{2\pi\delta t}\right)^{\frac{N}{2}} \left(\prod_{j=1}^{N-1}\int dq_j\right) \tag{8}$$

Proof. For some N, define $\delta t = T/N$ and $q_j = q(j\delta t)$.

$$
\begin{aligned}
\langle q_F|H|q_I\rangle =&(\prod_{j=1}^{N-1}\int dq_j)\prod_{i=0}^{N-1}\langle q_{j+1}|e^{-iHT}|q_j\rangle \\
=&(\prod_{j=1}^{N-1}\int dq_j)\prod_{i=0}^{N-1}\langle q_{j+1}|e^{-i\left(\frac{\hat{p}^2}{2m}+V(\hat{q})\right)\delta t}|q_j\rangle\text{, from (5)} \\
=&(\prod_{j=1}^{N-1}\int dq_j)\prod_{i=0}^{N-1}\langle q_{j+1}|e^{-i\frac{\hat{p}^2}{2m}\delta t}e^{-iV(\hat{q})\delta t}|q_j\rangle\text{, we'll expand in eigenvectors of }\hat{q} \\
=&(\prod_{j=1}^{N-1}\int dq_j)\prod_{i=0}^{N-1}\int dq\,\langle q_{j+1}|e^{-i\frac{\hat{p}^2}{2m}\delta t}|q\rangle\langle q|e^{-iV(\hat{q})\delta t}|q_j\rangle\text{, since }|q\rangle\langle q|=I
\end{aligned}
\tag{9}
$$

We will evaluate the integrand in two steps, starting with the potential energy.

$$
\begin{aligned}
\langle q|e^{-iV(\hat{q})\delta t}|q_j\rangle =&\langle e^{iV(\hat{q})\delta t}q|q_j\rangle \\
=&\langle e^{iV(q)\delta t}q|q_j\rangle\text{, since }|q\rangle\text{ is an eigenvector, with eigenvalue }q \\
=&\langle e^{iV(q)\delta t}q|q_j\rangle \\
=&e^{-iV(q)\delta t}\langle q|q_j\rangle
\end{aligned}
\tag{10}
$$

Now for the kinetic energy.

$$
\begin{aligned}
\langle q_{j+1}|e^{-i\frac{\hat{p}^2}{2m}\delta t}|q\rangle =&\int\frac{dp}{2\pi}\langle q_{j+1}|e^{-i\frac{\hat{p}^2}{2m}\delta t}|p\rangle\langle p|q\rangle \\
=&\int\frac{dp}{2\pi}e^{-i\frac{p^2}{2m}\delta t}\langle q_{j+1}|p\rangle\langle p|q\rangle\text{, since }|p\rangle\text{ is an eigenvector of }\hat{p} \\
=&\int\frac{dp}{2\pi}e^{-i\frac{p^2}{2m}\delta t}e^{ipq_{j+1}}e^{-ipq}
\end{aligned}
$$

Using (4) with $a=\frac{i\delta t}{m}$ and $J=q_{j+1}-q$:

$$
\begin{aligned}
\langle q_{j+1}|e^{-i\frac{\hat{p}^2}{2m}\delta t}|q\rangle =&\int\frac{dp}{2\pi}e^{\frac{-i\delta t}{2m}p^2}e^{ip[q_{j+1}-q]} \\
=&\frac{1}{2\pi}\left(\frac{2\pi m}{i\delta t}\right)^{\frac{1}{2}}e^{-\frac{m(q_{j+1}-q)^2}{2i\delta t}} \\
=&\left(\frac{-im}{2\pi\delta t}\right)^{\frac{1}{2}}e^{i\delta t\frac{m(q_{j+1}-q)^2}{2\delta t^2}}
\end{aligned}
\tag{11}
$$

Substituting (10) and (11) in (9):

$$\begin{aligned}
\langle q_F|H|q_I\rangle =&(\frac{-im}{2\pi\delta t})^{\frac{N}{2}}(\prod_{j=1}^{N-1}\int dq_j)\prod_{i=0}^{N-1}(\frac{-im}{2\pi\delta t})^{\frac{1}{2}}\int dq e^{i\delta t\frac{m(q_{j+1}-q)^2}{2\delta t^2}}e^{-iV(q)\delta t}\underbrace{\langle q|q_j\rangle}_{=m^2}\delta(q-q_j)\\
=&(\frac{-im}{2\pi\delta t})^{\frac{N}{2}}(\prod_{j=1}^{N-1}\int dq_j)\prod_{j=0}^{N-1}e^{i\delta t\left[\frac{m(q_{j+1}-q_j)^2}{2\delta t^2}-V(q_j)\right]}\\
=&(\frac{-im}{2\pi\delta t})^{\frac{N}{2}}(\prod_{j=1}^{N-1}\int dq_j)e^{i\sum_{i=0}^{N-1}\delta t\left[\frac{m(q_{j+1}-q_j)^2}{2\delta t^2}-V(q_j)\right]} \qquad (12)
\end{aligned}$$

Now, as $N\to\infty$

$$\sum_{i=0}^{N-1}\delta t\to\int dt$$

$$\frac{m(q_{j+1}-q_j)^2}{2\delta t^2}\to\frac{m\dot{q}}{2}\text{, so, substituting (6) and (8), (12) becomes}$$

$$\langle q_F|H|q_I\rangle\to\int Dq(t)e^{i\int_0^T dtL}\text{i.e. (7)}$$

□

Theorem 3.

$$\frac{\int_{-\infty}^{\infty}dxe^{-\frac{1}{2}ax^2}x^{2n}}{\int_{-\infty}^{\infty}dxe^{-\frac{1}{2}ax^2}}=\frac{\prod_{k=1}^{n}\left(2k-1\right)}{a^n}$$

Proof. The proof by mathematical induction starts by defining the proposition $P(n)$:

$$P(n)\equiv\left[\frac{\int_{-\infty}^{\infty}dxe^{-\frac{1}{2}ax^2}x^{2n}}{\int_{-\infty}^{\infty}dxe^{-\frac{1}{2}ax^2}}=\frac{\prod_{k=1}^{n}\left(2k-1\right)}{a^n}\right]$$

Lemma 4. *P(0) is true.*

Proof. $P(0)$ reduces to $\frac{\int_{-\infty}^{\infty}dxe^{-\frac{1}{2}ax^2}}{\int_{-\infty}^{\infty}dxe^{-\frac{1}{2}ax^2}}=1$ □

Lemma 5.

$$\forall\alpha>0,\ \frac{d}{d\alpha}\int_{-\infty}^{\infty}dxe^{-\frac{1}{2}\alpha x^2}x^{2n}=\int_{-\infty}^{\infty}dx\frac{\partial}{\partial\alpha}\left[e^{-\frac{1}{2}\alpha x^2}x^{2n}\right]$$

Proof.

$$\forall a > 0, \forall \alpha_0 \in (0, \alpha) \int_a^\infty dx e^{-\frac{1}{2}\alpha x^2} x^{2n} = \int_a^\infty dx e^{-\frac{1}{2}\alpha_0 x^2} e^{-\frac{1}{2}(\alpha-\alpha_0)x^2} x^{2n}$$
$$\text{Now define } M(x) = e^{-\frac{1}{2}\alpha_0 x^2}$$

We can choose a large enough that $e^{-\frac{1}{2}(\alpha-\alpha_0)x^2} x^{2n}$ is monotone decreasing for $x > a$. Clearly

$$\begin{aligned} e^{-\frac{1}{2}\alpha_0 x^2} e^{-\frac{1}{2}(\alpha-\alpha_0)x^2} x^{2n} &\in C \\ e^{-\frac{1}{2}\alpha_0 x^2} &\in C \\ e^{-\frac{1}{2}\alpha_0 x^2} e^{-\frac{1}{2}(\alpha-\alpha_0)x^2} x^{2n} &< M(x) \ \forall x > a \text{ and} \\ \int_a^\infty dx M(x) &< \infty \end{aligned}$$

for α in closed interval [A,B] that includes α_0; the Weierstrass M-Test [2, Chapter 10, 6.1] shows that the integral converges uniformly for $\alpha \in [A, B]$. Hence we can differentiate under the integral sign [2, Chapter 10, 8.3]. Now

$$\int_{-\infty}^\infty dx e^{-\frac{1}{2}\alpha x^2} x^{2n} = 2\int_a^\infty dx e^{-\frac{1}{2}\alpha x^2} x^{2n} + \int_{-a}^a dx e^{-\frac{1}{2}\alpha x^2} x^{2n}$$

Noting that $e^{-\frac{1}{2}\alpha x^2} x^{2n} \in C^1$ for x in [-a,a], we can also differentiate the second integral. □

Lemma 6. $P(0) \implies P(1)$

Proof. Since (2) converges uniformly, Lemma 5 allows us to differentiate under the integral sign[2].

$$\begin{aligned} -2\frac{d}{da}\int_{-\infty}^\infty dx e^{-\frac{1}{2}ax^2} &= -2\frac{d}{da}\left(\frac{2\pi}{a}\right)^{\frac{1}{2}} \\ \int_{-\infty}^\infty dx e^{-\frac{1}{2}ax^2} x^2 &= \left(\frac{2\pi}{a^3}\right)^{\frac{1}{2}} \\ &= \left(\frac{2\pi}{a}\right)^{\frac{1}{2}} \frac{1}{a} \\ &= \int_{-\infty}^\infty dx e^{-\frac{1}{2}ax^2} \frac{1}{a} \end{aligned}$$

□

Lemma 7. $P(n) \wedge P(1) \implies P(n+1)$

Proof.

$$
\begin{aligned}
P(n) \implies & \\
\int_{-\infty}^{\infty} dx e^{-\frac{1}{2}ax^2} x^{2n} &= \prod_{k=1}^{n} \big(2k-1\big) \int_{-\infty}^{\infty} dx e^{-\frac{1}{2}ax^2} \frac{1}{a^n} \\
-2\frac{d}{da} \int_{-\infty}^{\infty} dx e^{-\frac{1}{2}ax^2} x^{2n} &= -2 \prod_{k=1}^{n} \big(2k-1\big) \frac{d}{da} \big[\int_{-\infty}^{\infty} dx e^{-\frac{1}{2}ax^2} \frac{1}{a^n} \big] \\
\int_{-\infty}^{\infty} dx e^{-\frac{1}{2}ax^2} x^{2(n+1)} &= -2 \prod_{k=1}^{n} \big(2k-1\big) \Big[\int_{-\infty}^{\infty} dx e^{-\frac{1}{2}ax^2} (-\frac{1}{2}x^2) \frac{1}{a^n} - \int_{-\infty}^{\infty} dx e^{-\frac{1}{2}ax^2} \frac{n}{a^{n+1}} \Big] \\
&= \prod_{k=1}^{n} \big(2k-1\big) \Big[\underbrace{\int_{-\infty}^{\infty} dx e^{-\frac{1}{2}ax^2} x^2}_{\text{Now we apply } P(1)} \frac{1}{a^n} + 2 \int_{-\infty}^{\infty} dx e^{-\frac{1}{2}ax^2} \frac{n}{a^{n+1}} \Big] \\
&= \prod_{k=1}^{n} \big(2k-1\big) \Big[\int_{-\infty}^{\infty} dx e^{-\frac{1}{2}ax^2} \frac{1}{a} \frac{1}{a^n} + \int_{-\infty}^{\infty} dx e^{-\frac{1}{2}ax^2} \frac{2n}{a^{n+1}} \Big] \\
&= \prod_{k=1}^{n} \big(2k-1\big) \frac{2n+1}{a^{n+1}} \int_{-\infty}^{\infty} dx e^{-\frac{1}{2}ax^2} \\
&= \prod_{k=1}^{n+1} \big(2k-1\big) \frac{1}{a^{n+1}} \int_{-\infty}^{\infty} dx e^{-\frac{1}{2}ax^2} \\
&\equiv P(n+1)
\end{aligned}
$$

□

Summarizing:

$$
\begin{aligned}
\text{Lemma 4} &\implies P(0) \\
P(0) \wedge \text{Lemma 6} &\implies P(1) \\
P(0) \wedge P(1) \wedge P(n) \wedge \text{Lemma 7} &\implies P(n+1)
\end{aligned}
$$

□

Theorem 8. *If A is a symmetric matrix*

$$
\int_{-\infty}^{\infty} dx_1 dx_2 ... dx_N e^{-\frac{1}{2}\vec{x}A\vec{x}+\vec{J}\cdot\vec{x}} = \left(\frac{(2\pi)^N}{|A|} \right)^{\frac{1}{2}} e^{\frac{1}{2}\vec{J}\cdot A^{-1}\vec{J}} \tag{13}
$$

Proof. Since A is symmetric, there exists an orthogonal matrix O such that $A = O^T DO$, where D is a diagonal matrix–[3]. Define

$$
\begin{aligned}
y_i &= \sum_j O_{ij} x_j \\
\int_{-\infty}^{\infty} dx_1 dx_2 ... dx_N e^{-\frac{1}{2}\vec{x}A\vec{x}+\vec{J}\cdot\vec{x}} &= \left(\frac{1}{|O|}\right)^N \int_{-\infty}^{\infty} dy_1 dy_2 ... dy_N e^{-\frac{1}{2}\vec{x}O^T DO\vec{x}+\vec{J}\cdot\vec{x}} \\
&= \int_{-\infty}^{\infty} dy_1 dy_2 ... dy_N e^{-\frac{1}{2}(\vec{x}O^T)D(O\vec{x})+\vec{J}\cdot\vec{x}} \\
&= \int_{-\infty}^{\infty} dy_1 dy_2 ... dy_N e^{-\frac{1}{2}\vec{y}D\vec{y}+(O\vec{J})\cdot\vec{y}} \\
&= \prod_{i=1}^{N} \int_{-\infty}^{\infty} dy_i e^{-\frac{1}{2}D_{ii}y_i^2+(OJ)_i y_i} \\
&= \prod_{i=1}^{N} \left[\left(\frac{2\pi}{D_{ii}}\right)^{\frac{1}{2}} e^{\left(\frac{[(OJ)_i]^2}{2D_{ii}}\right)}\right] \qquad (14)
\end{aligned}
$$

But

$$
\begin{aligned}
\prod_{i=1}^{N} \left(\frac{2\pi}{D_{ii}}\right) &= \frac{(2\pi)^N}{|D|} \\
&= \frac{(2\pi)^N}{|A|}, \text{ since } O \text{ is orthogonal} \qquad (15)
\end{aligned}
$$

And

$$
\begin{aligned}
\prod_{i=1}^{N} e^{\left(\frac{[(OJ)_i]^2}{2D_{ii}}\right)} =& e^{\sum_{i=1}^{N}\left(\frac{[(OJ)_i]^2}{2D_{ii}}\right)} \\
\sum_{i=1}^{N}\left(\frac{[(OJ)_i]^2}{2D_{ii}}\right) =& \frac{1}{2}\sum_{i=1}^{N}(OJ)_i(D^{-1})_{ii}(OJ)_i \\
=& \frac{1}{2}\sum_{i,j,k,l=1}^{N} O_{ik}J_k(D^{-1})_{ij}O_{jl}J_l \\
=& \frac{1}{2}\sum_{i,j,k,l=1}^{N} J_k O^T_{ki}(D^{-1})_{ij}O_{jl}J_l \\
=& \frac{1}{2}\sum_{k,l=1}^{N} J_k\left(\sum_{k,l=1}^{N} O^T_{ki}(D^{-1})_{ij}O_{jl}\right)J_l \\
=& \frac{1}{2}\sum_{k,l=1}^{N} J_k\left(A^{-1}\right)_{kl}J_l \\
=& \frac{1}{2}JA^{-1}J
\end{aligned}
\tag{16}
$$

Substituting (15) and (16) in (14), we obtain (13). □

1.2 From Field to Particle to Force

$$
\begin{aligned}
W(J) =& -\frac{1}{2}\int\int d^4x d^4y J(x)D(x-y)J(y) \\
D(x-y) =& \int \frac{d^4k}{(2\pi)^4}\frac{e^{ik(x-y)}}{k^2-m^2+i\epsilon} \\
J(x) =& J_1(x)+J_2(x), \text{ where} \\
J_a(x) =& \delta^{(3)}(\vec{x}-\vec{x_a})
\end{aligned}
$$

Considering only the cross terms

$$
\begin{aligned}
W(J) =& -\int \frac{1}{(2\pi)^4} \cancel{2}\frac{1}{\cancel{2}} \int\int d^4x d^4y d^4k \delta^{(3)}(\vec{x}-\vec{x_1}) \frac{e^{ik(x-y)}}{k^2-m^2+i\epsilon} \delta^{(3)}(\vec{y}-\vec{x_2}))\\
=& -\frac{1}{(2\pi)^4} \int\int\int\int\int\int dx^0 d\vec{x} dy^0 d\vec{y} dk^0 d\vec{k} \delta^{(3)}(\vec{x}-\vec{x_1}) \frac{e^{ik^0(x^0-y^0)-i\vec{k}\cdot(\vec{x}-\vec{y})}}{k^2-m^2+i\epsilon} \delta^{(3)}(\vec{y}-\vec{x_2}))\\
=& -\frac{1}{(2\pi)^4} \int\int\int\int\int dx^0 d\vec{x} dy^0 dk^0 d\vec{k} \delta^{(3)}(\vec{x}-\vec{x_1}) \frac{e^{ik^0(x^0-y^0)-i\vec{k}\cdot(\vec{x}-\vec{x_2})}}{k^2-m^2+i\epsilon}\\
=& -\frac{1}{(2\pi)^4} \int\int\int\int dx^0 dy^0 dk^0 d\vec{k}\ \frac{e^{ik^0(x^0-y^0)-i\vec{k}\cdot(\vec{x_1}-\vec{x_2})}}{k^2-m^2+i\epsilon}\\
=& -\frac{1}{(2\pi)^3} \int\int dx^0 dy^0 \underbrace{\int \frac{dk^0}{2\pi} e^{ik^0(x^0-y^0)}}_{\delta(x^0-y^0)} \int d\vec{k}\ \frac{e^{i\vec{k}\cdot(\vec{x}-\vec{y})}}{k^2-m^2+i\epsilon}\\
=& -\frac{1}{(2\pi)^3} \int dx^0 \int d\vec{k}\ \frac{e^{i\vec{k}\cdot(\vec{x}-\vec{y})}}{k^2-m^2+i\epsilon}
\end{aligned}
$$

1.3 Coulomb and Newton

The Lagrangian Density is given by:

$$
\begin{aligned}
\mathcal{L} =& -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}m^2 A_\mu A^\mu + A_\mu J^\mu, \text{ where}\\
F_{\mu\nu} =& \partial_\mu A_\nu - \partial_\nu A_\mu \text{ and}\\
\partial_\mu J^\nu =& 0. \text{ We define the action}\\
S(A) =& \int d^4\mathcal{L}\\
=& \int d^4x \Big[-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}m^2 A_\mu A^\mu + A_\mu J^\mu \Big]
\end{aligned} \tag{17}
$$

Expanding the first term in the Lagrangian

$$
\begin{aligned}
F_{\mu\nu}F^{\mu\nu} =& g^{\mu\rho}g^{\nu\sigma}F_{\mu\nu}F_{\rho\sigma}\\
=& g^{\mu\rho}g^{\nu\sigma}\big(\partial_\mu A_\nu - \partial_\nu A_\mu\big)\big(\partial_\rho A_\sigma - \partial_\sigma A_\rho\big)\\
=& g^{\mu\rho}g^{\nu\sigma}\big[\partial_\mu A_\nu \partial_\rho A_\sigma - \partial_\mu A_\nu \partial_\sigma A_\rho - \partial_\nu A_\mu \partial_\rho A_\sigma + \partial_\nu A_\mu \partial_\sigma A_\rho\big]\\
=& g^{\mu\rho}g^{\nu\sigma}\partial_\mu A_\nu \partial_\rho A_\sigma - g^{\mu\rho}g^{\nu\sigma}\partial_\mu A_\nu \partial_\sigma A_\rho - \underbrace{g^{\mu\rho}g^{\nu\sigma}\partial_\nu A_\mu \partial_\rho A_\sigma}_{\text{substitute: } \mu\nu\rho\sigma \to \nu\mu\sigma\rho} + \underbrace{g^{\mu\rho}g^{\nu\sigma}\partial_\nu A_\mu \partial_\sigma A_\rho}_{\mu\nu\rho\sigma \to \nu\mu\sigma\rho}\\
=& g^{\mu\rho}g^{\nu\sigma}\partial_\mu A_\nu \partial_\rho A_\sigma - g^{\mu\rho}g^{\nu\sigma}\partial_\mu A_\nu \partial_\sigma A_\rho - g^{\nu\sigma}g^{\mu\rho}\partial_\mu A_\nu \partial_\sigma A_\rho + g^{\nu\sigma}g^{\mu\rho}\partial_\mu A_\nu \partial_\rho A_{ssigma}\\
=& 2\big[g^{\mu\rho}g^{\nu\sigma}\partial_\mu A_\nu \partial_\rho A_\sigma - g^{\mu\rho}g^{\nu\sigma}\partial_\mu A_\nu \partial_\sigma A_\rho\big]
\end{aligned}
$$

So the first term in the action is

$$-\frac{1}{4}\int d^4x F_{\mu\nu}F^{\mu\nu} = -\frac{2}{4}\int d^4x\big[g^{\mu\rho}g^{\nu\sigma}\partial_\mu A_\nu \partial_\rho A_\sigma - g^{\mu\rho}g^{\nu\sigma}\partial_\mu A_\nu \partial_\sigma A_\rho\big].$$ Integrating by parts gives

$$\begin{aligned}
&= \frac{1}{2}\int d^4x\big[g^{\mu\rho}g^{\nu\sigma}A_\nu \partial_\mu\partial_\rho A_\sigma - g^{\mu\rho}g^{\nu\sigma}A_\nu \partial_\mu\partial_\sigma A_\rho\big] \\
&= \frac{1}{2}\int d^4x\big[g^{\nu\sigma}A_\nu \partial^2 A_\sigma - A_\nu \partial^\rho\partial^\nu A_\rho\big] \\
&= \frac{1}{2}\int d^4x A_\nu\big[g^{\nu\sigma}\partial^2 - \partial^\sigma\partial^\nu\big]A_\sigma
\end{aligned}$$

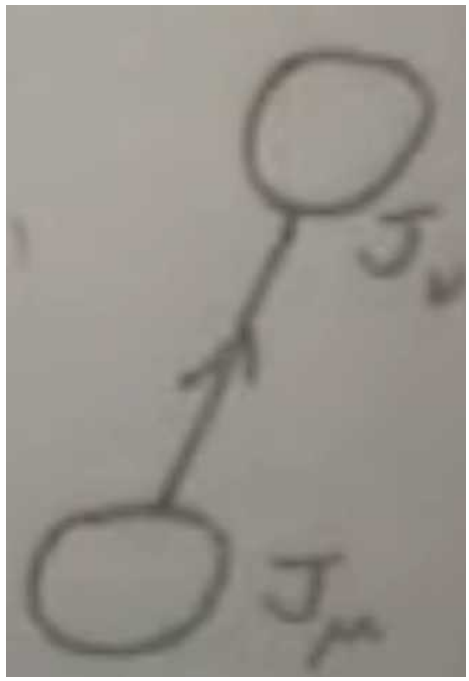
So the action becomes

$$\begin{aligned}
\int d^4\mathcal{L} &= \int d^4x\left[\frac{1}{2}A_\nu\big[g^{\nu\sigma}\partial^2 - \partial^\sigma\partial^\nu\big]A_\sigma + \frac{1}{2}m^2 A_\mu A^\mu + A_\mu J^\mu\right] \\
&= \int d^4x\left[\frac{1}{2}A_\nu\big[g^{\nu\sigma}(\partial^2 + m^2) - \partial^\sigma\partial^\nu\big]A_\sigma + A_\mu J^\mu\right]
\end{aligned}$$

$$D_{\mu\nu}\big(D^{-1}\big)^{\nu\lambda} = \delta_\mu^\lambda$$

From Lecture. How do we get a repulsive force? This is tied up with spin. If we want particles fro field to have spin 1, we need a vector–A_μ. We need to reduce A_μ to 3 degrees of freedom (polarization), and (17) is the only Lorentz invariant way to do this. We will start with the physics–Figure 1.

Figure 1: Electromagnetic Field with source and sink



$$
\begin{aligned}
J\phi \rightarrow & J_\mu A^\mu \\
\frac{1}{k^2+m^2} \rightarrow & \frac{1}{k^2+m^2} \underbrace{\underbrace{\sum_{a=1}^{3}}_{\text{3 polarizations}} \epsilon_\mu^{(a)}(k)\epsilon_\nu^{(a)}(k)}_{\text{A Lorentz tensor, } -G_{\mu\nu} \text{ say}}
\end{aligned} \tag{18}
$$

We will evaluate in the rest frame of the particle, where $k = (m, 0, 0, 0)$, and polarization is the 3 directions in Cartesian space.

$$
\begin{aligned}
\epsilon_\mu^{(1)}(k) =& (0, 1, 0, 0) \\
\epsilon_\mu^{(2)}(k) =& (0, 0, 1, 0) \\
\epsilon_\mu^{(3)}(k) =& (0, 0, 0, 1) \\
k^\mu \epsilon_\mu =& 0, \text{ since } k = (m, 0, 0, 0)
\end{aligned} \tag{19}
$$

(19) is Lorentz Invariant! c.f. $\partial_\mu A^\mu = 0$. We will establish the tensor $G_{\mu\nu}$ in (18) using symmetry and Lorentz invariance. There are only two things to

construct $G_{\mu\nu}$ from, k_μ and $g_{\mu\nu}$.

$$\begin{aligned}
-G_{\mu\nu} =& Ak_\mu k_\nu + Bg_{\mu\nu} \\
G_{\mu\nu}k^\mu =& 0 \text{ from (19), whence} \\
\big(Ak_\mu k_\nu + Bg_{\mu\nu}\big)k^\mu =& 0 \\
\big(A \underbrace{k^2}_{=m^2} + B\big)k_\nu =& 0 \\
A =& -\frac{B}{m^2} \\
-G_{\mu\nu} =& -\big(g_{\mu\nu} - \frac{k_\mu k_\nu}{m^2}\big). \text{ Normalization fixed by rest frame.} \\
\frac{1}{k^2+m^2} \to& \frac{-g_{\mu\nu} + \frac{k_\mu k_\nu}{m^2}}{k^2+m^2}
\end{aligned}$$

Imagine two stationary charges: then $J^i = 0$

$$\begin{aligned}
W(J) =& \int \frac{d^4k}{(2\pi)4} J^\mu(k)^* \frac{-g_{\mu\nu} + \frac{k_\mu k_\nu}{m^2}}{k^2+m^2} J^\nu(k) \\
\partial_\mu J^\mu =& 0 \text{ implies} \\
k_\mu J^\mu =& 0 \\
W(J) =& \int \frac{d^4k}{(2\pi)4} J^0(k)^* \frac{-g_{00}}{k^2+m^2} J^0(k) \\
=& \int \frac{d^4k}{(2\pi)4} J^0(k)^* \frac{-1}{k^2+m^2} J^0(k) \text{–Like charges repel}
\end{aligned}$$

Spin 2 has two indices $h_{\mu\nu}$. Two sign switches cancel!
Even spin can exchange lumps, but not odd.

Theorem 9.

$$\big(\delta^2 + m^2\big)A^\mu = 0 \textit{ and} \tag{20}$$

$$\partial_\mu A^\mu = 0 \textit{ are equivalent to} \tag{21}$$

$$\big(g^{\mu\nu}\partial^2 - \partial^\mu\partial^\nu\big)A_\nu + m^2 A^\mu = 0 \tag{22}$$

Proof: (20) $\wedge$ (21) $\implies$ (22). From (21)

$$\begin{aligned}
\partial^\nu A_\nu =& 0, \text{ whence} \\
\partial^\mu\partial^\nu A_\nu =& 0. \text{ Now (20) can be rearranged} \qquad (23) \\
\partial^2\big(g^{\mu\nu}A_\nu\big) + m^2 A^\mu =& 0 \\
g^{\mu\nu}\partial^2 A_\nu + m^2 A^\mu =& 0. \text{ Now, using (23)} \\
g^{\mu\nu}\partial^2 A_\nu - \partial^\mu\partial^\nu A_\nu + m^2 A^\mu =& 0, \text{ which rearranges to (22)}
\end{aligned}$$

□

(22) $\implies$ (20) $\wedge$ (21). From (22)

$$\begin{aligned}
m^2 A^\mu =& \big(\partial^\mu \partial^\nu - g^{\mu\nu}\partial^2\big) A_\nu \\
m^2 \partial_\mu A^\mu =& \partial_\mu \big(\partial^\mu \partial^\nu - g^{\mu\nu}\partial^2\big) A_\nu \\
=& \partial_\mu \partial^\mu \partial^\nu A_\nu - g^{\mu\nu}\partial_\mu \partial^2 A_\nu \\
=& \partial^2 \partial^\nu A_\nu - \partial^\nu \partial^2 A_\nu \\
=& 0, \text{ which is (21). Now (22) becomes} \\
g^{\mu\nu}\partial^2 A_\nu - \cancel{\partial^\mu \partial^\nu A_\nu} + m^2 A^\mu =& 0, \text{ which is (20)}
\end{aligned}$$

□

1.4 Feynman Diagrams

$$\begin{aligned}
Z(J,\lambda) =& \int D\phi \; e^{i\int d^4x \; \left[\frac{1}{2}[(\partial\phi)^2 - m^2\phi^2] - \frac{\lambda}{4!}\phi^4 + J\phi\right]} \\
=& \int D\phi \; e^{i\int d^4x \; \left[\frac{1}{2}[(\partial\phi)^2 - m^2\phi^2] + J\phi\right]} e^{-i\frac{\lambda}{4!}\phi^4}
\end{aligned}$$

The *functional derivative* is defined in [4]:

$$\begin{aligned}
\int \frac{\delta F(x)}{\delta \rho}\phi(x)dx \triangleq& \lim_{\epsilon\to 0} \frac{F(\rho+\epsilon\phi) - f[\rho]}{\epsilon} \\
=& \left[\frac{dF(\rho+\epsilon\phi)}{d\rho}\right]_{\epsilon\,=\,0}
\end{aligned}$$

Theorem 10.

$$Z(J,\lambda) = e^{-\frac{i\lambda}{4!}\int dx \frac{\delta^4}{(\delta J)^4}} Z(J,0)$$

Proof. After [5]... □

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